

PianoHandSync: An Alignment-based Hand Pose Discrepancy Visualization System for Piano Learning

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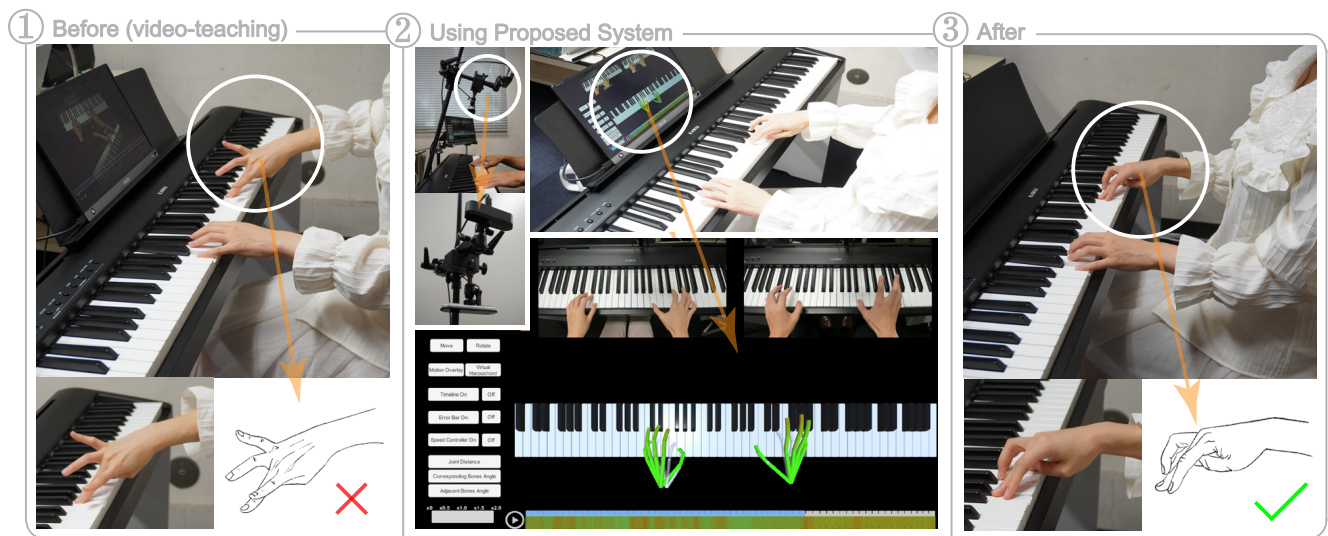


Figure 1: PianoHandSync explores discrepancy feedback and visualization design through aligned performances to improve the user experience of correcting their finger postures in piano learning.

ABSTRACT

Video-based lessons are becoming a popular way for distance piano education. However, limited by the fixed camera angle, a video is difficult to tell precise 3D hand posture, which is one of the most essential factors for learning piano. This paper presents a visualization system providing the intuitive discrepancy of hand postures in two piano performance videos. Through a motion capture system, the estimated 3D postures are visualized and discrepancies based on distinct metrics are displayed, integrated with modular functions assisting skill acquisition. A pilot study proves that the proposed visualization can be a supplementary means for only video-based lessons in terms of correcting hand postures and fingering.

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585705>

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Piano Training, Motion Analysis, Virtual Visualization

ACM Reference Format:

Ruofan Liu, Erwin Wu, Chen-Chieh Liao, Hayato Nishioka, Shinichi Furuya, and Hideki Koike. 2023. PianoHandSync: An Alignment-based Hand Pose Discrepancy Visualization System for Piano Learning. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585705>

1 INTRODUCTION

Nowadays, online piano lessons are gaining popularity thanks to the development of multimedia platforms. On the other hand, correct postures are one of the most essential factors for playing the piano. In real-world lessons, students are asked to recognize their mistakes

by visual cues and direct feedback from the instructor, then try to understand and mimic an expert's pose. However, this becomes difficult for video teaching because of the fixed camera perspectives in 2D videos and the lack of teacher feedback.

From our initial interviews with some professional pianists, we find that even a pro also has these concerns when playing in a fast rhythm. That is the reason why professional pianists also record their performances at different speeds and review them in their practice. Many sports or instrument training systems obtain users' postures and give feedback to them in real time. However, this is technically not possible for rhythmical skills like piano since the play speed between novice and expert is different, of which the posture cannot be compared frame-by-frame.

Therefore, in this paper, we present an alignment-based piano self-training system that reconstructs 3D representations from piano videos and provides visual feedback on fingering differences between the videos of a user and a teacher. The proposed system consists of three main modules: a pose estimation module, a pose alignment module, and a feedback visualization module (as shown in Figure 2).

To better understand the user experience of our system as well as the training effect, we conducted a pilot study with ten piano novices and compared it to conventional learning methods using video-teaching instruction. The study includes qualitative questionnaires and quantitative performance evaluations. From the final results, despite that the proposed method has several system errors such as the error in pose estimation and in pose alignment, it suggests that our system provides a better understanding than a video-teaching method and shows significant improvements in some designed tasks.

In summary, our key contributions are as follows: We propose the first alignment-based motion difference feedback method for providing the discrepancies between recorded and tutorial data for piano learning. A prototype system is implemented for piano learning and a specific hand pose estimation model with a temporal pose alignment algorithm is developed for comparing piano hand postures. A pilot study is conducted with both quantitative and qualitative evaluations and several questionnaires which suggests the proposed system has a better training effect over conventional video-teaching methods. A comprehensive discussion of the pros and cons of the current prototype is conducted, followed by an elaboration of possible future modifications.

2 RELATED WORK

In this section, we describe assisted technologies for piano training and discuss visualization feedback in piano learning.

2.1 Assisted Technologies for Piano Training

There are a large number of interactive assistive technologies to help learn the piano. For example, new keyboards emerged to better distribute feedback and assist pianists. McPherson et al. [11] presented an alternative approach to keyboard controller design, which measured five distinct gestural dimensions. Apart from this, researchers have developed other music keyboards reporting additional information for each key-press, involving continuous key

position sensing [4], keys moving in two dimensions, and feature touch sensitivity on the surface [12].

Directly reorganizing the piano keyboard is a time-consuming and labor-intensive task. Therefore, Takegawa et al. [19] projected a piano roll on the keyboard. However, the authors had to paint the entire keyboard white and the projector required complex calibration, which inflicts other demands. Other researchers used projection systems and feedback for note correctness and accuracy to reduce cognitive load during piano learning [15, 16].

Much of the previous research has dealt with the mechanical finger movements in piano practice-assisted systems. Mobile Music Touch (MMT) [6] is a lightweight, wireless haptic music instruction system, which demonstrates the benefits of haptic sensations in enhancing motor sequence recall. Based on this, Takahashi et al. [18] proposed a soft exoskeleton glove with human anatomical architecture to facilitate fine control of sequential force production. But in the above systems, the predefined finger sequences place the student in a passive learning environment.

2.2 Visualization for Piano Acquisition

Previous studies suggested that motion analysis in a virtual environment could assist in skill acquisition, hence applied in various sports. Nozawa et al. [14] presented a VR ski training system, visualizing prerecorded teacher's motion in different ways for users to imitate. Based on this, there're other detailed visualization methods, for example, time distortion can affect sports training, in kendama [7], in tennis [5], and many other sports training [9, 10, 21].

Other than providing visual feedback for varied sports, researchers have explored diverse feedback approaches for piano learning as well. Wang [20] proposed PPVR, a VR collaboration system designed to support remote piano training from different angles. Based on this, HoloKeys [17], with an augmented reality user interface and a piano playing apparatus, aimed to extend online music education beyond desktop platforms into the physical world. In addition to the VR and AR, Liu et al. [8] proposed a piano training application on a tablet device, focusing on the hand pose discrepancies. But they didn't conduct a detailed user study when comparing baselines and visualizations.

In general, all of the above methods have limitations to being used in practical applications due to their precondition, additional equipment, or lack of user study. To the best of our knowledge, there are few approaches to recreating the playback of finger movements or comparing hand posture from two performances. In contrast, our proposed system can record users' performances and display the discrepancies between their hands and the teacher's in 3D visualization. Moreover, it can be used not only on touch screens but also in virtual reality environments.

3 IMPLEMENTATION

3.1 Apparatus and Hand Tracking

We used the Kawai ES110 portable digital piano and Roland UM-ONE to record the midi data. For hand motion capture, we employed a Logitech Brio web camera and a synchronized IR-based depth camera Ultra 3Di. Lastly, we used a tablet, ASUS ZenScreen MB16AMT-J touch-type USB portable LCD display, to avoid the load of keyboard

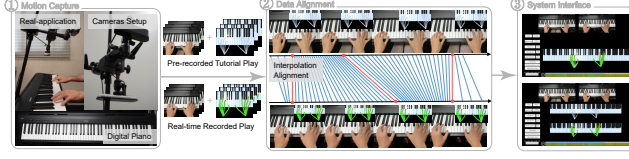


Figure 2: System architecture diagram. A complete visual presentation consists of three modules: a pose estimation module (motion capture), a pose alignment module (data alignment), and a feedback visualization module (system interface).

and mouse operation. Besides, we filmed participants’ finger movements during the whole performing practice via a side camera.

To realize 3D visualization, the 3D hand poses need to be extracted from the 2D videos. We utilize a computer vision-based hand pose estimation model, InterHand [13], which can precisely estimate 2D and 3D dual hand poses from a single webcam using deep neural networks. Since the 3D pose estimation model is relatively heavy (approx. 10-FPS in our setup), it may be sufficient for extracting from online videos but not suitable for real-time training. Therefore, for the later pilot study, we calibrate another IR depth camera with the webcam to perform camera triangulation. From the web camera images, we first estimate the 2D hand joints position using a trained model of [13] (which runs in roughly 30-FPS) and then obtain the corresponding 3D coordinates from the depth camera to recover a 3D hand posture.

3.2 Alignment Algorithm

To better discover where the student’s hands differed from the teacher’s during the whole performance, it’s indispensable to analyze the positions and angles of their fingers at each unit moment. However, there is no guarantee that the user will always play with the exact same rhythm as the teacher, which makes it enormously difficult to find the discrepancies. Therefore, it is essential to align the moments when the two performers are supposed to show the same hand posture. Without a doubt, it should be the frame when a specific key is either pressed or released.

On the other hand, MIDI data recording any key pressed or released at any moment can make it actualized. Once there are specific source time stamps for every tone of the two compared performances, these keyframes are then used to align the moments when an exact note is played and the segments between the two adjacent notes through linear interpolation. Thus, after calculating source time stamps into several embedding spaces, combining these interpolation spaces together makes an aligned moments sequence following the teacher’s.

3.3 Discrepancy Metrics

To measure the differences between the recorded (student) and tutorial (teacher) data more comprehensively and holistically, at every frame, we have three discrepancy metrics, Corresponding Joint Interval (CJI), Corresponding Finger Slope (CFS), and Adjacent Bone Angle (ABA), respectively.

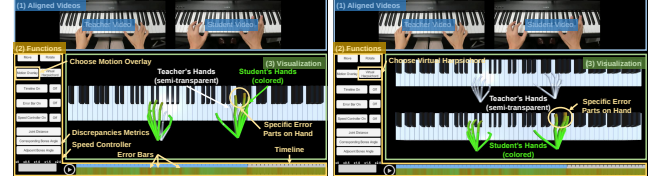


Figure 3: System Interface. Left: Motion Overlay. Right: Virtual Harpsichord.

CJI. CJI denotes the value of distances from every joint of the student to that of the teacher. Each hand has 20 joints in total (4 joints: index, middle, ring, pinky; 3 joints: thumb; 1 joint: wrist). So every frame gives 40 CJI ($0 \leq i \leq 39$).

CFS. CFS stands for the angle discrepancy between the corresponding bone slope of the teacher and student. Extensively, the index, middle, ring, and pinky fingers all have distal, middle, proximal phalanx, and metacarpal bones, while the thumb doesn’t have a middle bone, hence each hand has 19 bones in total. Every frame gives 38 CFS ($0 \leq i \leq 37$).

ABA. Based on the finger slope, the discrepancy between the angles formed by adjacent bones is raised to conduct a more comprehensive measurement. Every two bones constitute such one angle, (three for index, middle, ring, and pinky, two for thumb, and four formed by nearby metacarpal bones), hence each frame gives 36 ABA in total ($0 \leq i \leq 35$).

3.4 Visualization interface

As Figure 3 shows, the system interface has three panels: (1) Aligned Videos. (2) Functions. (3) Visualization. In the aligned videos, the left is from the teacher’s raw data, while the right is the student’s after being processed and aligned with the teacher. The other two panels, functions and visualization, will be described in detail in the following sections.

Modular Functions. Aimed to evaluate the preference for various visualization functions, we integrate them into a modular interface. All auxiliary function controls are located on the left side of the main interface, and tools including timeline, error bars, and speed controller can be either hidden or activated. The timeline and speed controller are similar to that in video-teaching, moving the timeline cursor to decide when to start replay and the speed ranges from 0 to 2 times. Besides, a series of error bars will be shown on the timeline, where an error bar represents one error moment. We use it to emphasize the mistake-made frames in the whole performance and indicate where and what specified kinds of errors are located. From the three discrepancy metrics (defined in Section 3.3), we will obtain three corresponding error bar sequences. It’s allowed to either show all of these error sequences or merely exhibit one of them on the timeline.

Discrepancy Display. In sum, our system has to display the discrepancies between student and teacher’s performance in two aspects, i.e. on particular hands visualization and the error bars. In the previous research [8], three definitive thresholds were arranged to decide the discrepancy display color. In our system, we used a Gaussian

filter to make color changes continuously. We only deposit the maximum thresholds respectively for three different metrics. Color will gradually change from green to red, representing changes from no discrepancy to the maximum threshold we set beforehand. In terms of hand visualization, it is clear that all the discrepant areas should be explicit, so the color of all parts (including joints and bones) changes, ranging from green to red.

Virtual Keyboard Emplacement. For the sake of visual clarity, we straight-forwardly place the student's hand and teacher's hand in a motion-overlay manner and distinguish their two hands by distinct color, almost transparent grey (teacher's) and colorful (student's), so that the discrepancies are intuitive and easily apprehended. On the other hand, based on our previous pilot study [8], some professional pianists suggest they appeal for a side-by-side keyboard-placing mode to check and compare the two performances individually, rather than the above kind of "noisy" overlaid visualization. Thus we develop another emplacement approach, to display the four hands on two separate keyboards, which is inspired by a harpsichord (an instrument with two or more sets of strings).

4 PILOT STUDY

Being the first hand-pose-discrepancy-based feedback system for piano learning, we compared our system with an existing video teaching method to understand the user experience, limitations, and insights into improving the feedback design. We recruited ten participants (8 M, 2 F) for a within-subjects (repeated-measures) study design, from a local institution. All of the participants had zero experience in piano performances in the past, and thus classified as beginners.

4.1 Conditions and Tasks

Conditions. Each participant was asked to complete some tasks in two distinct teaching conditions: (1) **Video-teaching**, only online video teaching, in which the tutorial teacher's playing video was played back over and over again, both pausing and choosing where to start replaying were permitted, while the user would learn and practice freely; and, (2) **Visualization-system**, using our visualization system, in which the integrated functions and discrepancy feedback would support user's learning and exercising.

Tasks. We designed two simple tasks, and both of them were possible for beginners to learn within the study time considered by experienced pianists, and the data (i.e. videos and 3D hand data) used for tutorials are from professionals (> 12 years of piano experience). (1) **Task 1**, task 1 was a music scale, a set of musical notes ordered by fundamental frequency and pitch, involving an ascending scale and a descending scale. Only the thumb and pinky of both hands were used in this task (C, C, D, D, E, E, F, F, G, G, A, A, B, B, C, C, C, B, B, A, A, G, G, F, F, E, E, D, D, C, C). (2) **Task 2**, task 2 was an arpeggio, the notes composing a chord played in a rising or descending order. This task was the arpeggio for the chord of C major going up two octaves and then going back, i.e. the notes would be (C, E, G, C, E, G, C).

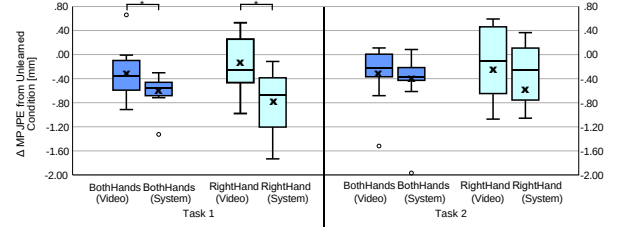


Figure 4: Positional Improvement Score (PIS) of two tasks: MPJPE reduction from the unlearned condition. Left: Task1; Right: Task2.

4.2 Procedure

To better examine the use of the proposed system, we designed the procedure as follows:

- (1) The participants receive an introduction and become familiar with our system to reduce the learning effect. And they have 3 minutes to watch a tutorial video and perform a warm-up before each task.
- (2) The initial performance is recorded three times for comparison.
- (3) The participants start a training session in one of the above-mentioned conditions (Video/Visualization) for 10 minutes. After each training, the performance is recorded again for three repetitions.*
- (4) The procedure (2)-(3) is repeated for the two conditions, while the order is shuffled for each participant using the Balanced Latin Square [3] to counter-balance the learning rate.
- (5) The procedure (2)-(4) is repeated for both tasks, which results in 2 conditions $\times$ 2 tasks = 4 training sessions.
- (6) Finally, after all training, a SUS [1] questionnaire and a customized questionnaire about auxiliary functions in a 7-Point Likert-Scale are given to the participants. One-to-one detailed interviews are also conducted.

* In the Visualization, the participants are free to update their posture while the loading time is not counted in the 10-minute training.

5 RESULTS

5.1 Positional Improvement Score

Positional Improvement Score (PIS) denotes the change volume from the unlearned state, i.e. the disparity of Mean Per Joint Position Error (MPJPE) between the moment just completing the 3-minutes tutorial and the moment after a teaching condition experience. We use MPJPE to estimate the similarity between the user's finger postures and the teacher's. Therefore, the smaller PIS is, the more similar the user's hand pose to the teacher's data.

The left side of Figure 4 summarizes the specific PIS made by users after each condition in task 1, the decrease of the visualization system is significantly much more vs. only the video-teaching condition (independent t-test, $p < 0.05$). In detail, we found that using the visualization system had statistically significantly ($t(18) = 2.083$, $p = 0.044$) lower MPJPE ($PIS = \Delta MPJPE = -0.69 \pm 0.48mm$) at the end of every 10-minutes piano performance practice compared to the video-teaching method ($PIS = \Delta MPJPE = -0.34 \pm 0.55mm$).

when calculating both hands. For right-hand-only, there's also significant ($t(18)=2.751$, $p=0.013$) improvement between the visualization and the video-teaching.

The right side of Figure 4 shows the PIS made in task 2, the decrease of the visualization system is a little bigger than the video-teaching condition. But from the independent t-test, we don't find significant improvement. In detail, we found that using the visualization system had lower MPJPE ($PIS = \Delta MPJPE = -0.46 \pm 0.57mm$) compared to the video-teaching method ($PIS = \Delta MPJPE = -0.31 \pm 0.49mm$).

5.2 SUS and Functions Experience

The System Usability Scale (SUS) was used to perceive usability, and the average score (at the 50th percentile) is 68. The score calculated from our questionnaire was 72.75, which was above average and acceptable. The score of modular functions, (i.e. move and rotate, timeline, error bars, speed controller, discrepancy metric) are 4.80, 6.50, 6.10, 5.70, and 4.60. As for the keyboard placement, there's a significant ($t(18)=2.125$, $p=0.048$) difference between motion-overlay (5.90) and virtual-harpsichord (4.30).

6 DISCUSSIONS

6.1 Piano Performance and Interview

Our visualization system demonstrated some improvements in piano learning from quantitative and qualitative results.

Our main finding was that this visualization way results in more correct and effective practicing of finger-refined movements. As professional pianists suggest, this is critical to piano performance, especially when considerable movement control is required to accomplish high levels of timing and force precision. Almost none of these are currently possible with only conventional video-learning methods.

After the quantitative evaluation, we conducted detailed interviews with each user. The users were asked about whether this system can help piano video teaching and their thoughts on system improvement. 9 out of 10 users think that visualization can be more intuitive to see discrepancies, especially postures and fingering, which cannot be straightforwardly accomplished in video-learning. Therefore, this 3D visualization is a very good auxiliary tool for traditional videos to help users quickly and accurately locate errors.

6.2 Function Availability

As is mentioned in Section 5.2, timeline, error bars and speed controller gain the highest acceptance rate due to their capability to clearly point out the mistake-made moments. This indicates that the majority of users regarding the discrepancies provisioning as a satisfying visualization approach. It helps them quickly and efficiently locate mistakes and greatly reduces the mental burden of self-observation and reflection in the traditional video-teaching method.

Three discrepancy metrics are diverse among the users because this time the users are piano novices, which means that they have little experience playing the piano and hence do not have a basic concept of fingering and proper hand posture. It will inevitably lead to the fact that they will one-sidedly apprehend and thus fail to utilize the discrepancy metrics comprehensively. However, the

function of this system to display diverse types of errors may be more suitable for professional pianists.

When comparing the motion overlay (single keyboard) and virtual harpsichord (separate keyboards) method, more users prefer the former. We hypothesize it's because the motion overlay is more intuitive and easy for beginners to find the discrepancy when the hands of students and teachers are overlaid. But for pros, results may be different in our future user study. Regardless of what the visualization interface is like, the users all gave higher ratings and preferences.

6.3 Future Work

User Study for Professional Pianists. This time we conducted a pilot study with novices to test the system's usability and acquire improvement opinions. In actuality, the target users are reputed to be professional pianists. To settle this issue, we want to first improve the experience of the interactive operations based on this-time preliminary study, then we have partnered with Sony Computer Science Laboratories and intend to contact at least 14 pros to participate in our future user study.

More Accurate Alignment. In our study this time, we used a linear interpolation algorithm to align the keyframes between recorded and tutorial data. It's unequivocal the most appropriate with the least calculation time consumed which is proper for a real-time system. But this time the tasks are designed more simply to fit the beginners, the fingering and movements between each keyframe are uncomplicated. When considering more complex tasks for expert pianists, we demand more accurate alignment results, not only every key-pressed moment but also every fine finger movement made during the moment. Temporal Cycle-Consistency Learning (TCC) is a potential solution [2], which has been certificated to successfully align videos by simply matching frames using the nearest neighbors in the learned embedding space. We want to modify self-supervised TCC into a slightly innovative and different network architecture supervised by keystroke data in our future work, instead of only aligning the key-pressed moment. Using the "supervised TCC", we can have key-pressed moment-aligned sequences as well as the key movements associated.

7 CONCLUSION

In this paper, we show a workflow of how visualization for piano learning is designed and evaluated. We recorded the hand postures through motion capture and studied different features in the 3D hand visualization system which can assist learning, such as modular functions, discrepancy display, and keyboard emplacement. The system was then studied in a pilot experiment with 10 novices. As a result, the proposed system outperforms the video-teaching condition in the learning effect in hand postures. And users can better observe and comprehend the discrepancies using the visualization system. Overall, in the conducted evaluations, the positive feedback from users has already suggested that the presented approach provides a suitable complementary option to current methods in piano learning. We also believe that the visualization methods and discrepancy displays used here could provide some insight for other skill acquisition studies.

ACKNOWLEDGMENTS

This work is supported by JST CREST, under Grant No.: JPMJCR17A3.

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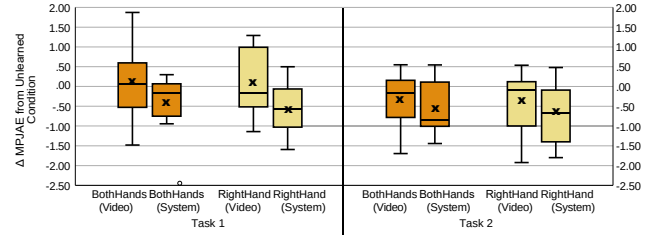


Figure 5: Angular Improvement Score (AIS) of two tasks: MP-JAE reduction from unlearned condition. Left: Task1; Right: Task2.

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A OTHER QUANTITATIVE RESULTS

A.0.1 AIS. Angular Improvement Score (AIS), which actually corresponds to the discrepancy metric CFS and ABA (in Section 3.4), denotes the change volume from unlearned state, i.e. the disparity of Mean Per Joint Angle Error (MPJAE) between the moment just completing the 3-minutes tutorial and the moment after a teaching condition experience. We use MPJAE to estimate the angular score for users' finger postures. Therefore, the smaller AIS is, the more similar the joint angle to the teacher's data the user showed.

The left side of Figure 5 shows the AIS made in task 1, the decrease of the visualization system is a little bigger than only video-teaching condition. But from the independent t-test, we don't find significant improvement. In detail, we found that using the visualization system had lower MPJAE ($AIS = -0.46 \pm 0.82$) of both hands compared to video-teaching method ($AIS = 0.0540 \pm 0.95$). As for only the MPJAE of the right hands, the system also had lower value ($AIS = -0.59 \pm 0.67$) than video-teaching method ($AIS = 0.053 \pm 0.81$).

The right side of Figure 5 shows the AIS made in task 2, the decrease of the visualization system is a little bigger than the video-teaching condition. But from the independent t-test, we don't find significant improvement. In detail, we found that using the visualization system had lower MPJAE ($AIS = -0.56 \pm 0.69$) of both hands when compared to video-teaching method ($AIS = -0.33 \pm 0.72$). As for only the MPJAE of the right hands, the system also had lower value ($AIS = -0.60 \pm 0.78$) than video-teaching method ($AIS = -0.34 \pm 0.77$).

A.0.2 NASA-TLX. According to the Shapiro-Wilk Normality test, all measures except *Effort* were not normally distributed. As such, every measure was analyzed using Friedman analyses. Because there're only two different conditions, video-teaching and visualization-system, Wilcoxon signed-rank tests with Holm-Bonferonni corrections for post-hoc comparisons are not indispensable. Besides, for all TLX measures, median values are reported. As Figure 6 shows, in

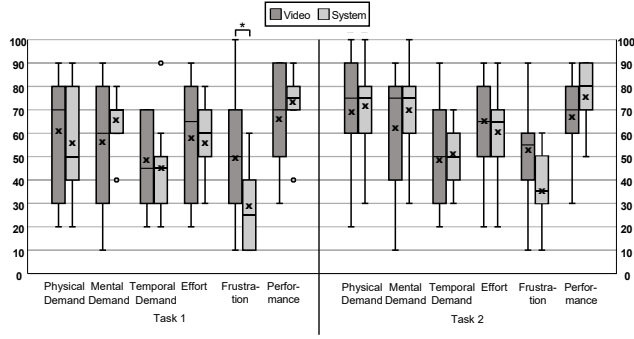


Figure 6: NASA-TLX Ratings. Lower values correspond to lower mental, physical, and temporal demand, as well as lower effort, lower frustration, but worse performance.

general, we found that SYSTEM outperformed VIDEO in FRUSTRATION for both two tasks. However, for other measures (i.e. PHYSICAL DEMAND, MENTAL DEMAND, TEMPORAL DEMAND, EFFORT, and PERFORMANCE), there is no obvious difference between the two conditions. As for FRUSTRATION, SYSTEM was less frustrating in both two tasks. A Friedman analysis found a significant effect on Frustration in task 1 ($\chi^2(2) = 7.00, p = 0.008$) and task 2 ($\chi^2(2) = 2.78, p = 0.096$) depending on which type of teaching condition was conducted (i.e. video-teaching vs. visualization-system).

Respectively, in task 1, SYSTEM is (29.00 ± 17.00), and VIDEO is (49.00 ± 26.63). In task 2, SYSTEM is (35.00 ± 16.40), and VIDEO is (52.00 ± 24.82).

Physical Demand: For both two tasks, there was no significant difference in participant ratings between VIDEO and SYSTEM. In task 1, SYSTEM is (56.00 ± 22.89), and VIDEO is (61.00 ± 26.25). In task 2, SYSTEM is (72.00 ± 19.90), and VIDEO is (69.00 ± 25.87).

Mental Demand: Again, there was no significant difference in participant ratings between VIDEO and SYSTEM for both two tasks. In task 1, SYSTEM is (66.00 ± 11.14), and VIDEO is (56.00 ± 25.77). In task 2, SYSTEM is (70.00 ± 19.49), and VIDEO is (62.00 ± 25.22).

Temporal Demand: Once more, there was no significant difference in participant ratings between VIDEO and SYSTEM for both two tasks. In task 1, SYSTEM is (45.00 ± 19.10), and VIDEO is (49.00 ± 18.68). In task 2, SYSTEM is (51.00 ± 13.75), and VIDEO is (49.00 ± 21.19).

Effort: Continuing the trend, there was no significant difference in participant ratings between VIDEO and SYSTEM for both two tasks. In task 1, SYSTEM is (56.00 ± 15.62), and VIDEO is (58.00 ± 25.22). In task 2, SYSTEM is (60.00 ± 20.49), and VIDEO is (64.00 ± 21.54).

Performance: There were no differences between VIDEO and SYSTEM for either TASK. In task 1, SYSTEM is (70.00 ± 18.97), and VIDEO is (66.00 ± 20.10). In task 2, SYSTEM is (76.00 ± 12.81), and VIDEO is (67.00 ± 15.52).